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Enhanced Fault Classification, Detection, and Location Estimation in the IEEE 14-Bus Smart Grid Using a Hybrid CNN-LSTM Algorithm with Adaptive Learning Rate

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Abstract

Smart grids play a crucial role in ensuring reliable and sustainable energy delivery. However, the increasing complexity of transmission and distribution networks, driven by rising energy demands, presents significant challenges in fault detection, classification, and localization. Accurate and timely identification of faults both symmetrical and asymmetrical is essential to mitigate the long-time power outages and maintain grid stability when isolating the faulty part. This study proposes an advanced fault diagnosis approach using a hybrid convolutional neural network (CNN) and long short-term memory (LSTM) model. Currents and voltages data were collected from an Institute of Electrical and Electronics Engineers IEEE 14-bus test system, with various fault scenarios simulated in MATLAB Simulink. Without additional preprocessing, the raw of three-phase voltages and currents signals were directly input into a cascade-parallel reconstructed CNN utilizing varying filter lengths to capture diverse signal features. The extracted multi-scale features were then processed by the LSTM network to learn time-related dependencies effectively. The proposed hybrid CNN-LSTM methodology outperforms conventional approaches in fault analysis, achieving high accuracy across various fault diagnosis tasks. The study introduces three distinct classifiers: fault class type Identification, achieving 99.98% accuracy, line faulty section identification, with an accuracy of 96%, and fault location estimation, with 99.98% accuracy. The model's effectiveness is further validated using key performance metrics, including accuracy, precision, recall, and F1-score, demonstrating its robustness and reliability for real-time smart grid fault analysis.

Keywords Smart grids · Fault detection and classification · Deep learning · CNN · LSTM · IEEE 14-bus

1 Introduction

Fault detection and classification are critical components of power transmission line protection. Over the years, extensive research has been conducted to develop fast and accurate methods for detecting and classifying faults in transmission lines, ensuring the protection of the system from potential damage caused by faults. Additionally, accurate fault detection and classification play a vital role in fault location, significantly reducing fault clearing time and enhancing the overall reliability of the power transmission network.

The integration of distributed generators (DGs) into the distribution network offers several advantages, including reduced environmental pollution, enhanced power generation efficiency, and minimized transmission losses. Furthermore, DG integration improves voltage profiles, ensuring a higher-quality power supply for consumers. Compared to conventional power generation methods that rely on large-scale generators, DGs have a smaller size, lower production capacity, and reduced generation costs, making them a more cost-effective and sustainable alternative.

Smart grid technology has been introduced to mitigate the problem of renewable integration and improve the reliability of the system. Renewable generation is usually located near the load center. However, in the present system, the

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rapid growth of renewable energy has caused overloading in the distribution line due to reverse power flow. In DG protection using an overcurrent relay, it cannot differentiate between a fault and normal load switching. This can cause disruptions in service. Therefore, to improve the reliability of power distribution, DG needs appropriate protection. With the implementation of intelligent systems in the power system, it can provide a clear picture of the system status and is ideal for facilitating automation. This system status is essential for DG protection to make the decision whether to disconnect the fault [1–3].

Recent advances in modern electrical systems have opened a new era where utilities are considering the use of intelligent systems to improve the operation and utilization of the electrical system. By using intelligent systems, it will improve the reliability, economics, and quality of electrical power. The term “Smart Grid” has been used to refer to the modernization of the utility power system. The smart grid is an electric system that is based on two-way communications and makes use of information and communication technology. In general, the smart grid is an enabling infrastructure for the integration of renewable energy and the utilization of two-way communication [3, 4].

The increase in the electrical grid complexity has motivated the development of new methodologies for operating, monitoring, and controlling while seeking greater efficiency and robustness of electric power systems. They are not only formed by new operational schemes and different energy generation sources but have their optimization that is based on the use of information technology [5, 6]. As a result of the large amount of data generated mainly by new measurement technologies, classical methodologies based on static monitoring become obsolete. Analyzing the data through dynamic techniques based on artificial intelligence is standard in the electricity market. Furthermore, analyzing data through dynamic techniques based on artificial intelligence nowadays is standard in the electricity market. Moreover, these dynamic techniques play an essential role in transforming smart grid systems into self-healing systems that can diagnose the disturbance and recover the system rapidly, leading to a decrease in financial losses and maintenance time. [7–9].

Faults in electrical systems are unavoidable occurrences. Under normal conditions, the power system remains balanced; however, the occurrence of faults disrupts this balance. A fault can be defined as any significant abnormality in voltage or current within a power system, indicating a disturbance within the system. Transmission lines experience the highest incidence of faults, accounting for approximately 80% of power system faults due to their location, which is outside of the building [10].

Transmission line faults can generally be classified into two categories: symmetrical (balanced) and unsymmetrical (unbalanced) faults. Unsymmetrical faults can further be divided into single-line-to-ground faults (LG) and multiphases faults which are line-to-line (LL), line-to-line-to-ground (LLG), and line-to-line-to-line (LLL) or three-phase faults. The occurrence percentages for different types of faults in the system are as follows: Single-line-to-ground (LG) faults are the most common, occurring in 70–74% of cases. line-to-line (L-L) faults occur with a frequency of 12–15%, while double-line-to-ground (LL-G) faults account for 16–18% of occurrences. Lastly, three-phase (LLL) faults are the least frequent, comprising only 2–4% of all faults [11].

Transmission line protective relaying typically includes three different categories: fault detection, classification, and localization. Efficient fault detection ensures the rapid identification and isolation of faulty lines or sections, minimizing the potential for extensive system damage. Furthermore, precise fault classification is crucial for knowing the main cause and the severity of the fault, while fault location is important for identifying the location of the fault for rapid maintenance of the system, thereby reducing fault clearing time and speeding up the restoration of power service to the consumers [12].

Over the past years, significant research has focused on fault classification in various sections of the power system, particularly within transmission and distribution networks. One approach involves impedance measurement techniques. While this technique is simple, time efficient, and relies on the fundamental frequency harmonics of voltage and current, it suffers from low precision when fault resistance values are high [13].

More recent research efforts have shifted toward machine learning techniques, which employ various algorithms designed with ability of learning and extracting insights features from input data without the need for specific programming for different fault types [14]. With large volumes of data available for training, machine learning methods are well suited to addressing the inherent variability of faults in power systems. Machine learning models can be trained to analyze complex relationships among fault characteristics, providing valuable insights that may not be readily apparent. This approach offers advantages of high precision and time efficiency in fault detection and classification while this suffers from time consumption, complexity, and need for a large amount of data [15].

In recent years, numerous researchers have applied machine learning and feature engineering techniques in electrical systems for fault detection and classification. Various approaches have been proposed, including those utilizing wavelet transform (WT) as a powerful tool for feature extraction, artificial neural networks (ANN), and fuzzy systems, each contributing to advancements in fault identification and categorization. In order to identify and categorize faults in overhead transmission lines, wavelet transform (WT) was used as a feature extraction method in both the frequency and time domains [16]. Many researchers aim to combine multiple algorithms to achieve more accurate classifiers with the reducing of the computation time. Despite their achievement, they applied their proposed algorithm to a simple network, which can face different scenarios when applied to a complicated network. Furthermore, the noise effects were not considered. Additionally, the study avoids mentioning the impact of using different levels of wavelet transformation.

A combination of discrete wavelet transform (DWT) along with fuzzy logic algorithms, employing DWT as a feature extraction to extract the important features and fuzzy logic for the classification task [17]. Additionally, a fuzzy logic algorithm combined with WT has been developed for identifying various fault types within electrical distribution systems [18]. These studies work perfectly when the noise is simple and has a small variation while the complexity of this algorithm leads to time-consuming and computational burden.

Artificial neural networks (ANN) in a combination with the wavelet transform (WT) have been utilized in transmission lines for fault detection, classification, and location. Several ANN-based methods have been proposed for transmission line protection, leveraging the ability of artificial neural networks (ANN) to be trained on offline data, which is beneficial for power system applications. Although the study proves its immunity to the effect of the high fault resistance, it does not provide an in-depth analysis of the sudden load change as it reminds challenges because it can be considered a fault. An investigation into the use of ANN was presented for fault detection based on voltage and current signatures [19]. Although the use of neural networks enhances fault diagnosis, it still presents difficulties like computing complexity and detection precision. Furthermore, the use of the artificial neural network uses time-series data of voltages, currents, or both, which lacks the frequency domain. As a result, it leads to the inadequacy of the features that were used to distinguish between classes.

For the diagnosis of PV array faults, a unique deep learning-based method was introduced; nevertheless, it has certain drawbacks [20]. One notable drawback is its moderate accuracy with the presence of noise. However, the proposed CNN-based method achieves 70.45% accuracy, which, although an improvement over previous techniques, may still be insufficient for real-world deployment where noise is a common challenge in PV system data. Also, an extended ANN applications was presented for fault type and location identification [21]. The proposed technique presents an innovative deep learning-based approach for fault detection in smart grids, it still possesses limitations. They conducted high accuracy rates, while the study lacks analysis of performance under noise and the network complexity. Furthermore, the Clarke–Concordia technique concentrates on identifying the type and location of faults in dual transmission lines [22]. However, the proposed algorithm has not been tested in the presence of noise. Furthermore, the proposed technique was not applied on different fault scenarios, such as high-impedance faults or the scalability of larger network.

Further research, an article presents an effective method for fault detection and isolation in DC microgrids by utilizing artificial neural networks (ANNs) and wavelet transform [23]. The study does not evaluate the method's performance under various fault types or network configurations, like high-impedance faults or multi-fault conditions. Furthermore, the scalability of the approach to larger, more complex DC networks and their computational requirements were not discussed. On the other hand, another study utilizes ANN to determine fault types and locations by analyzing voltage and current signals in IEEE 14-bus test network [24]. While the study presents a new technique for fault detection, classification, and localization in transmission networks, there are some notable flaws. The paper does not discuss how the proposed method performs under various fault scenarios, including high-impedance faults or transient faults that may be more challenging to detect the fault. Moreover, the computational complexity of the proposed approach was not addressed, particularly for large-scale transmission networks. The use of artificial neural networks (ANNs) also presented for both classify and locate faults in transmission lines [25]. The study lacks computational efficacy and comparison with existing fault classification and location methods, which could help evaluate the relative strengths and weaknesses of the proposed technique.

Deep learning (DL) is a powerful algorithm that has gained wide acceptance in the field of artificial intelligence in recent years. The deep learning algorithm will be an excellent fit for extremely complex, layered, and intertwined challenge attributes. Deep learning has significantly advanced fault diagnosis in electric power systems, particularly as these systems grow increasingly complex. The need for advanced technologies capable of automatically extracting multilevel features from extensive datasets has become critical. As an end-to-end approach, deep learning surpasses traditional models by efficiently processing large datasets and eliminating the reliance on manually extracted features. Therefore, articles [26, 27] the authors employed an autoencoder with unsupervised learning and automatic feature extraction techniques for fault classification and detection in transmission lines. This study utilizes another facet of artificial intelligence and improves the fault diagnosis while still facing the lack of generalizability and computational burden, beside the need for large data.

In addition, articles [28–30] explored the application of deep belief networks (DBN) and proposed several promising new models that have recently emerged for faults detecting and classifying in transmission lines. Therefore, the study introduces an improved deep belief network (DBN) for fault classification in power transmission lines under different noise levels. However, the model shows robustness against noise; it does not consider the effects of varying fault resistances or network topology changes. Also, convolutional neural networks (CNNs) were employed as a classifier to identify different types of faults in an IEEE 30-bus system [31]. It was utilizing voltage and current signals from all buses as input without any additional preprocessing.

Furthermore, a CNN-based method for fault detection and classification in the IEEE 34-bus distribution system is also presented by the authors in [32]. By leveraging CNNs' capability to automatically extract multilevel features from smart meter data, the method eliminates the need for manual feature extraction. The proposed model achieves high accuracy and fast response times in fault detection. However, the study does not address fault location or faulty section identification, which are important for comprehensive fault management. On the other hand, a modified convolutional neural network (CNN) was presented for fault classification in distributed networks with distributed generators (DGs) [33]. The model directly processes raw three-phase voltage and current signals from various faults and no-fault scenarios. A cross-validation methodology is employed to evaluate the model's performance using different key metrics. However, the study demonstrated the effectiveness of the proposed method but still has limitations to adopt in large-scale electrical networks.

Consequently, in [21, 34], the authors introduce a customized CNNs for fault detection, classification, and localization in transmission and distribution networks incorporating DG resources. The proposed model is applied to IEEE (9-bus and 6-bus) test systems with consideration given to the effects of network topology changes. Although the CNNs possess a strong capability for extracting spatial features, their convolutional layers are less effective at capturing complex and variable temporal information. In contrast, LSTMs are highly effective in characterizing the temporal correlations or long dependencies inherent in time-series data.

In order to fill the gap in the current deep learning applications for power systems, the authors in [35] suggest a long short-term memory (LSTM)-based method for fault classification in transmission lines. The study models a 400 kV, 100 km transmission line and trains the LSTM network using fault-free and ten fault types, incorporating noise levels of 20 dB and 30 dB SNR to assess robustness. The proposed model achieves high classification accuracy, outperforming previous methods. However, the study does not explore the large-scale power grids. Additionally, while the method performs well under noise conditions, it does not assess performance under varying fault resistances or diverse network topologies, which are necessary factors for practical deployment.

A DWT-LSTM model for fault detection in insulated conductors within transmission line networks was proposed by different researchers. However, the model is limited in its ability to effectively extract spatial features. To overcome this limitation, many researchers have suggested combining CNNs with LSTM models to enhance the extraction of both spatial and temporal features [36]. Moreover, another study presents a hybrid CNN-LSTM deep learning model for sentiment analysis on Twitter datasets. The performance of the proposed model is compared with several other machine learning classifiers to evaluate its effectiveness [37].

Furthermore, the CNN and a weight dropped LSTM (WDLSTM) network were combined to create a hybrid deep learning model that provides efficient network intrusion detection system (IDS) [38]. CNN is used to extract relevant features from large IDS data, while the WDLSTM network captures long-term dependencies between these features, effectively mitigating overfitting. Consequently, different articles have proposed combining CNN and LSTM models to address the challenge of spatial-temporal feature extraction. A multi-channel CNN-LSTM (MCCNN-LSTM) network for fault detection and classification using the ENET dataset. The study employs fast Fourier transform (FFT) for preprocessing the three-phase voltage signals to extract both low- and high-frequency components. These components, along with the original signals, are fed into three parallel CNNs with varying filter lengths. An LSTM network is then utilized to sequentially integrate these multi-scale features, enhancing the model's ability to capture both spatial and temporal patterns in the data [39].

Enhanced wavelet techniques (scaled continuous wavelet transform, or S-CWT) to create scalogram images was used to investigate transmission line faults when combined with CNN. Although these studies have improved accuracy, noise immunity, and taken into account the majority of fault scenarios, they were not tested on ring-type networks or even when there was a sudden change in load [40, 41].

Despite all limitations and drawbacks mentioned above, this paper proposes an effective hybrid model combining parallel CNN and multiple LSTM cells for fault detection, classification, and localization in IEEE 14-bus test system in power system. The model is designed to automatically extract and analyze time sequences of three-phase voltage and current data collected from all buses, without the need for additional preprocessing. To capture deeper feature information within the signals, three-phase data is used as an input into multiple parallel CNNs with varying filter sizes, followed by a cascade of LSTMs to capture long-term temporal dependencies.

Unlike conventional CNN and LSTM models, the proposed hybrid CNN-LSTM approach can extract multi-angle and multi-directional spatial fault features. This enables a more detailed signal representation and comprehensive extraction of hidden features. Specifically, the proposed method consists of three classifiers: the first detects the faulty section, the second identifies the fault type, and the third estimates the fault location based on the dataset labels.

Furthermore, the results from testing showing significantly improved accuracy. This work illustrates the potential of deep learning for real-time diagnostics, monitoring, and operations in next generation of smart grids, aligning with advancements in technology and the integration of comprehensive smart grid applications in real-world settings.

The structure of this paper is organized as follows: Section 2 provides an overview of the studied system and the data generation process. Section 3 outlines the methodology and architecture of the proposed hybrid deep learning model. Section 4 presents the simulation results and offers an in-depth analysis of the findings. Furthermore, a detailed discussion of the proposed model and its comparison with previous studies is presented in Section 5. Finally, Section 6 concludes the paper by summarizing the key insights and contributions.

2 Data Generation and System Overview

2.1 System Studied

IEEE 14-bus system is a widely recognized benchmark in electrical engineering, extensively used for evaluating power system analysis, fault detection, and control algorithms. This test system provides a simplified realistic representation of the American Electric Power system, comprising 14 buses, 5 generators, 11 loads, and a network of 17 transmission lines. To facilitate fault detection and classification studies, a detailed mathematical framework of the IEEE 14-bus system was constructed in MATLAB. This framework serves as the foundation for simulating of various operational scenarios and generating the necessary data for model development and testing. The system's structure and components make it an ideal test case for analyzing the effectiveness of advanced fault detection and classification methodologies.

In this study, the impact of renewable distributed generation (DG) on fault detection and classification is analyzed by integrating two DG sources into the IEEE 14-bus system. A 20-MW photovoltaic (PV) source is strategically placed near the load at bus 13, and a 30-MW wind turbine is connected at bus 14. These additions introduce dynamic variations in power flow and fault characteristics, enabling a comprehensive evaluation of the proposed fault classification model under realistic operating conditions. Figure 1 illustrates the single-line diagram of the modified IEEE 14-bus system with the incorporated DG sources.

To generate the dataset that is required for training and evaluating the proposed model, a comprehensive set of fault and non-fault scenarios is simulated by systematically adjusting system parameters and settings. These include variations

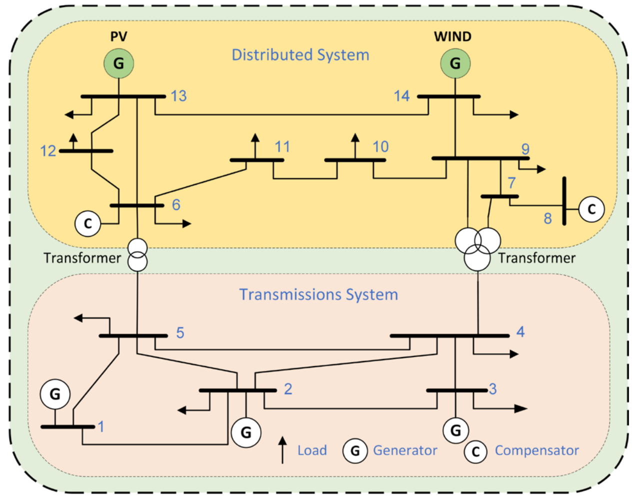

Fig. 1 Single-line diagram of IEEE 14-bus system

in fault types, locations, resistances, and faulty section. Table 1 outlines the detailed configurations for all simulated scenarios, covering both faulty and normal operating conditions. These diverse simulations are designed to ensure the robustness and versatility of the model in accurately detecting and classifying various fault types that may occur within a smart grid system.

The simulation data comprised three-phase voltage and current measurements collected from all buses, which were used to develop three classifiers: fault class type (FCT), line faulty detector (LFD), and fault location estimator (FLE). The data collection process incorporates a range of fault resistance values, systematically categorized as either low or high. Fault resistance, along with inception angle, represents a critical parameter in fault detection within electrical systems. Fault resistance variations are significantly influencing the voltage and current characteristics in the affected sections during fault condition, thereby playing a vital role in fault diagnosis.

The fault simulation in MATLAB was conducted using fault block from the Simulink library, which is a powerful tool for accurately modeling and analyzing faults in power systems. In this study, the fault block was configured to simulate various types of faults, including single-line-to-ground (L-G), double-line-to-ground (L-L-G), double-line (L-L), and three-phase (L-L-L) faults. These fault scenarios were applied at multiple locations across all transmission lines in the IEEE 14-bus test system.

The simulation setup included defining the fault parameters such as fault resistance, fault duration, and fault inception angle. The fault block was strategically placed within the Simulink model to ensure a realistic emulation of fault conditions. Each fault type was simulated at different percentages of the line length (e.g., 10%, 20%, up to 90%) to assess the impact of fault location variability on the test system. Additionally, the system parameters, such as bus voltages and line currents, were monitored and recorded for every simulation scenario to generate a comprehensive dataset.

The generated fault dataset forms the basis for training and validating the proposed hybrid deep learning model, ensuring its robustness and reliability across a comprehensive range of fault conditions. By encompassing diverse fault types, locations, and scenarios, the simulation delivers an in-depth analysis of fault dynamics within the IEEE 14-bus test system, thereby enhancing the model's capability to accurately detect, classify, and localize faults.

Therefore, incorporating varying fault resistance values while data collection is crucial for improving the accuracy of fault detection models and to include all possible behavior of the system. Table 2 presents the number of cases and corresponding labels for each class. The dataset is organized as follows: the fault class includes 5,600 cases, generated by modifying system parameters and configurations, while the non-fault class consists of 1,331 cases. For LFD classifier, the dataset totals 6,120 cases, and for FLE model, it comprises 6,300 cases. The MATLAB simulation model of the IEEE 14-bus system was used to collect data for each fault type, with proper labeling provided as shown in Table 2.

Figure 2 provides a detailed illustration of the interconnection of various power system components within the MATLAB simulation model of the IEEE 14-bus test system. The diagram highlights the placement of key elements such as generators, transformers, transmission lines, loads, and distributed generators (DGs). Notably, distributed generators are strategically integrated into the system, with one DG connected to Bus-13 and another to Bus-14. These DGs simulate realistic scenarios of renewable energy integration, such as solar or wind power, enhancing the complexity of the test system. The figures also showcases the connectivity between buses, emphasizing the power flow and the interaction of system components under normal and faulty conditions. This comprehensive model serves as the foundation for conducting fault analysis and evaluating the performance of the proposed fault detection and classification methodology.

2.2 Data Preprocessing

The data preprocessing phase begins with organizing and structuring the simulation data for efficient analysis and model training. The collected dataset is stored in a structured format as a CSV file within a Data Frame, containing key variables such as fault type, faulty line, fault location, and three-phase voltage and current measurements from all buses. The fault type variable categorizes faults into ten distinct classes: AG, BG, CG, AB, AC, BC, ABG, ACG, BCG, and ABCG. The faulty line variable identifies the specific transmission line where the fault occurs, ranging from line 1 to line 17. Additionally, the fault location parameter defines the fault's position along each line, spanning from 10% to 90% of the total line length. This structured data organization ensures consistency and enables efficient preprocessing for seamless model integration.

The next step in the preprocessing pipeline is data normalization, a critical technique that ensures all input features is on a uniform scale, thereby improving model performance and training stability. In this study, we employ min-max normalization, which rescales voltage and current signals to a standardized range, preventing features with inherently larger numerical values from dominating the learning process. Normalization is particularly important for deep learning models like CNN and LSTM, as it enhances convergence speed, reduces training time, and mitigates the risk of numerical instability. The normalization process transforms each feature by adjusting its scale relative to the minimum and maximum values in the dataset. The transformation is given by Equation 1:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{1}$$

where X is a random feature value that is to be normalized, $X_{\min}$ is the minimum feature value in the dataset, and $X_{\max}$ is the maximum feature value.

This transformation ensures that all feature values fall within a predefined range, typically [0,1] or [-1,1], depending on the activation functions used in the neural network. Standardizing voltage and current signals ensures that each feature contributes equally to model training, preventing bias toward attributes with naturally larger magnitudes. Additionally, normalization helps improve generalization by reducing the risk of overfitting. By ensuring consistent feature scaling, the model can effectively learn fault-related patterns across different operating conditions, leading to more robust and accurate fault classification.

3 Proposed Hybrid Deep Learning Methodology and Architecture

The methodology section provides a thorough explanation of the architecture of the proposed hybrid deep learning algorithm. The subsequent sections offer a detailed description of the architectural design and the mathematical framework that underlay the proposed approach:

3.1 CNN Architecture

CNN is a deep learning model particularly well suited for processing data with a grid-like topology, where time-series data represents a one-dimensional grid topology [42]. The CNN architecture, consisting of stacked convolutional and pool-

Table 1 Parameters and configurations for fault classification and detection

Parameters	Possible configuration	No. of cases
Fault class type (FCT)	AG, BG, CG, AB, BC, AC, ABG, BCG, ACG, ABC	10
Line faulty detector (LFD)	Line1-2, Line1-5, Line2-4, Line2-3, Line2-5, Line3-4, Line4-5, Line7-8, Line9-7, Line10-9, Line11-10, Line11-6, Line12-6, Line12-13, Line13-6, Line13-14, Line14-9	17
Fault location estimator (FLE)	10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%	9
Fault resistance	0.1, 0.5, 2.5, 10, 20, 50, 75, 100	8

Table 2 Data distribution for LFD, FCT, and FLE classifiers

LFD	No. of distribution	FCT	No. of distribution	FLE	No. of distribution
No-fault	1331	No-fault	1331	No-fault	1331
Line1-2	360	AG	560	10% LineLength	700
Line1-5	360	BG	560	20% LineLength	700
Line2-4	360	CG	560	30% LineLength	700
Line2-3	360	AB	560	40% LineLength	700
Line2-5	360	AC	560	50% LineLength	700
Line3-4	360	BC	560	60% LineLength	700
Line4-5	360	AB	560	70% LineLength	700
Line7-8	360	ACG	560	80% LineLength	700
Line9-7	360	BCG	560	90% LineLength	700
Line10-9	360	ABC	560		
Line11-10	360				
Line11-6	360				
Line12-6	360				
Line12-13	360				
Line13-6	360				
Line13-14	360				
Line14-9	360				

ing layers, facilitates the automatic extraction of underlying spatial features from raw data, allowing for a deep learning approach to capture complex patterns.

In the convolutional layer, feature maps from the previous layer are convolved with a 1D or 2D kernel, and the resulting feature maps are activated by an activation function. This one-dimensional CNN architecture, including convolutional layer, pooling layer, and fully connected layers, is particularly effective for detailed analysis of time-series data. The mathematical representation of a convolutional layer is expressed in Equation 2:

$$x_j^l = f\left(\sum_{i \in M_j} x_i^{l-1} * k_{ij}^l + b_j^l\right) \tag{2}$$

where

- x_j^l is the output of the j-th filter in the l-th layer.
- f is the activation function, such as ReLU or Sigmoid.
- $x_i^{(l-1)}$ is the input feature map from the previous $(l-1)$-th layer.
- k_{ij}^l is the kernel (or filter) weight connecting the i-th input feature to the j-th output feature in layer l.
- b_j^l is the bias term for the j-th filter in layer l.
- M_j represents the set of input features connected to the j-th output feature map.

The equation illustrates the process of calculating each output feature in a convolutional layer by applying a kernel to a specific subset of input features. The result is then passed through an activation function, effectively capturing spatial or temporal dependencies inherent in the input data. The development of CNNs has led to the emergence of various advanced architectures, including multi-channel and multi-

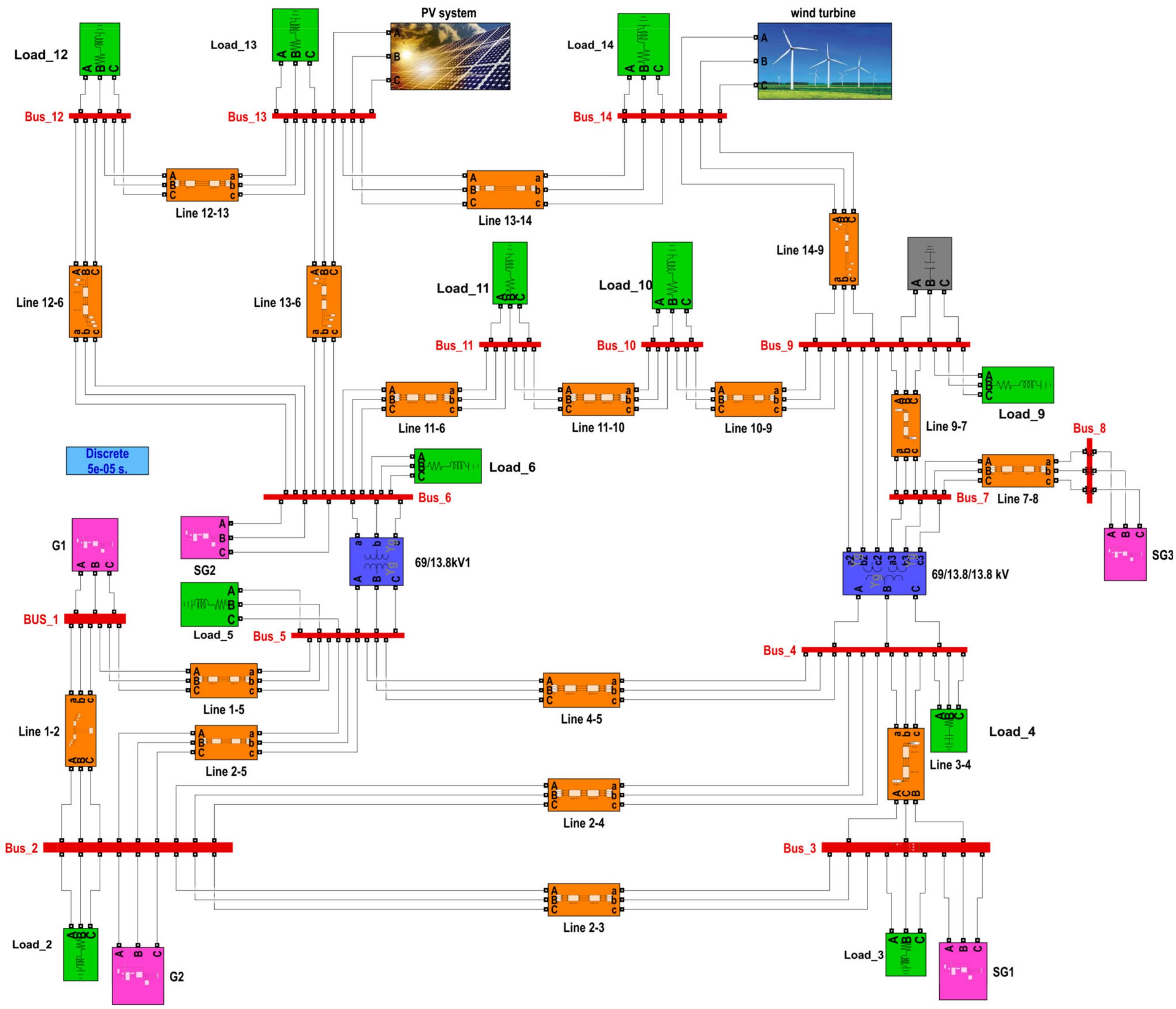

Fig. 2 MATLAB Representation and interconnection of the IEEE 14-Bus Test System

headed models. Despite the diversity in structures, most CNN architectures share a common foundational design.

A crucial operation in CNNs is the pooling layer, which helps reduce the spatial dimensions of feature maps while preserving the most important information. Pooling is typically achieved through subsampling [43], with common methods including average pooling, maximum pooling, and stochastic pooling. In our model, we utilize maximum pooling, as it selects the highest value from each region of the feature map, thereby emphasizing the most prominent and informative features in the data. This method aids in retaining critical information while reducing computational complexity.

3.2 LSTM Layers Architectures

The LSTM network is an advanced architecture within the recurrent neural network (RNN) family, developed to overcome the challenges faced by traditional RNNs, particularly the problems of vanishing and exploding gradients. These issues hinder the ability of traditional RNNs to effectively capture long-term dependencies in sequential data. LSTM networks address this by incorporating memory cells and gating mechanisms, which allow them to maintain and update information over long sequences, making them particularly well suited for tasks involving time-series data and sequential patterns.

The fundamental operation of the LSTM model is similar to that of a traditional RNN; however, LSTM introduces specialized gating mechanisms that significantly improve its

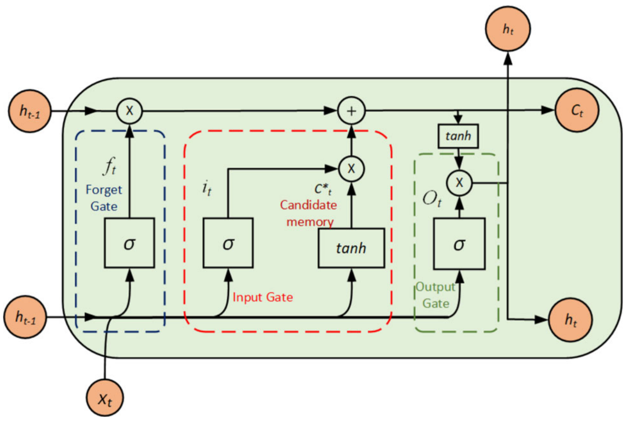

Fig. 3 LSTM Cell Architecture and Components

ability to retain and manage information over long sequences. These gates: the forget gate, input gate, and output gate, work together with an internal memory cell to regulate the flow of information, allowing the network to selectively store, forget, and output information as needed.

As illustrated in Figure 3, the LSTM network consists of three main gates: input gate i_t , forget gate f_t, and output gate O_t alongside a memory cell that stores essential information over time. The mathematical representation of each gate is provided as follows:

$$\left\{\begin{array}{l} i_t = \text{sigmoid}\left(W_i\left[h_{(t-1)}, X_t\right] + b_i\right) \\ f_t = \text{sigmoid}\left(W_f\left[h_{(t-1)}, X_t\right] + b_f\right) \\ C_t^* = \tanh\left(W_c\left[h_{(t-1)}, X_t\right] + b_c\right) \\ o_t = \text{sigmoid}\left(W_o\left[h_{(t-1)}, X_t\right] + b_o\right) \end{array}\right\} \quad (3)$$

where :

- X_t represents the input at time step t,
- h_{t-1} refer to the state output from the previous time step $t-1$,
- W_i, W_f, W_o represent the weight matrices for the input gate, forget gate, and output gate, respectively,
- b_i, b_f, b_o are the biases for the input gate, forget gate, and output gate, respectively.

The final output of an LSTM network, denoted as h_t, is determined by combining the outputs of the output gate O_t and the intermediate memory cell state C_t at a given time step t. This process allows the network to effectively capture both short-term and long-term dependencies. The computation for h_t is carried out as in Equation 4:

$$\left\{\begin{array}{l} C_t = C_{(t-1)} \cdot f_t + C_t^* \cdot i_t \\ h_t = O_t * \tanh\left(C_t\right) \end{array}\right\} \quad (4)$$

In LSTM network, the forget gate decides which information from the memory cell should be discarded, ensuring that irrelevant data does not clutter the memory. The input gate, on the other hand, determines what new information should be added to the memory cell. Finally, the output gate regulates how much of the internal memory cell's information is released as the network's output. This gating mechanism allows the LSTM to selectively forget unnecessary details while retaining crucial information, enabling the effective preservation and utilization of long-term dependencies in the data.

3.3 Hybrid CNN-LSTM Layer Architecture

Combining deep learning algorithms is a common strategy to improve model performance by addressing the limitations of individual methods. Although CNNs excel at extracting spatial features, they often struggle to capture long-term dependencies and are susceptible to overfitting, particularly in fully connected layers when handling high-dimensional data. The proposed hybrid CNN-LSTM structure, which is depicted in Fig. 4, overcomes these challenges by integrating an LSTM layer before the fully connected layers. However, the proposed approach leverages the strength of CNNs in spatial feature extraction and the capability of LSTMs to capture long-term dependencies, enabling superior feature extraction and classification. This integration effectively manages long-term dependencies, reduces overfitting, and enhances overall training accuracy. Furthermore, it demonstrates clear advantages over using CNN or LSTM models individually.

In this study, CNN architecture is employed to extract hidden features from raw three-phase voltage and current data, structured as a one-dimensional vector. The extracted features serve as input to an LSTM architecture, which identifies sequential dependencies within the data. The CNN component consists of three convolutional layers, each followed by a max-pooling layer, after which the output is flattened. The first convolutional layer processes the input sequence and projects it onto feature maps. The subsequent layers further process these features maps to refine and amplify significant patterns, with kernel size set at three-time steps to capture fine-grained local patterns and with progressively decreasing feature map counts (128, 64, and 32). The max-pooling layer simplifies these feature maps by retaining 20% of the prominent signal values [44, 45].

The flattened feature maps are then passed to the LSTM decoder, which includes three hidden layers with 100, 50, and 30 units, respectively. The LSTM reads the sequence, and its output is a vector that captures the underlying relationships in the data. The internal representation of the input sequence is repeated across all time steps in the output sequence, providing input to the LSTM model at each step. A fully connected layer interprets each time step in the output, ensuring that

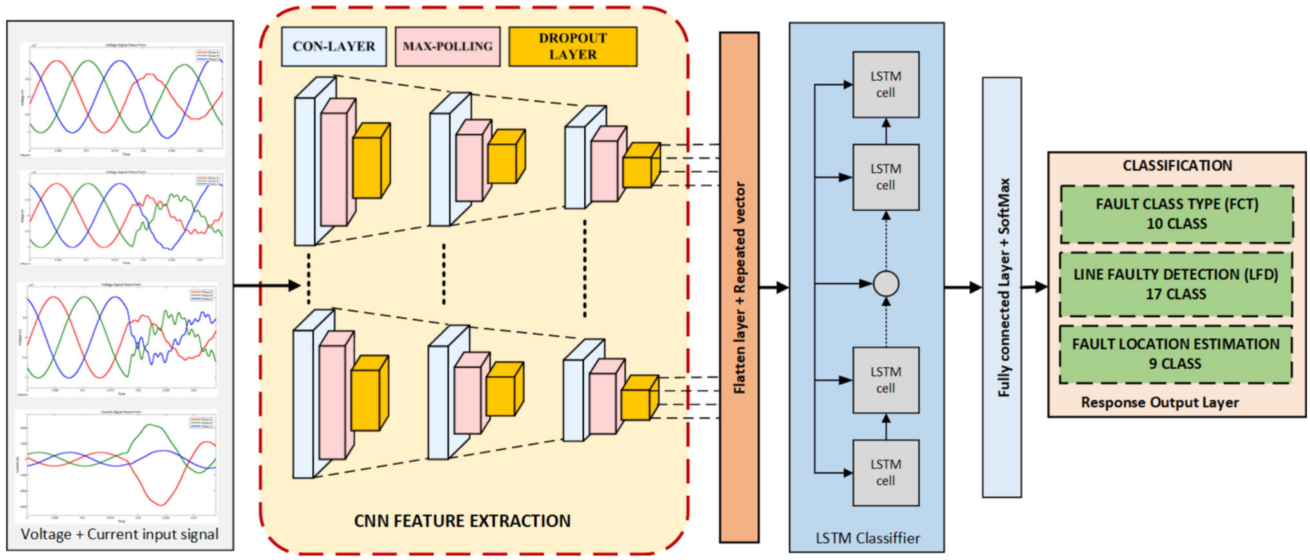

Fig. 4 LSTM cell architecture and components

Table 3 Hyperparameter configuration during training progress

Hyperparameter	Proposed CNN-LSTM
Epoch No	100
Batch size	(64-128)
Optimizer	Adam
Initial learning rate	0.01
Learning rate adjustment Drop=0.1, epoch_drop=10	Monitor = Validation loss, patience=10, epoch(drop*epochs/epoch_drop)
Dropout	(0.2−0.25)
Activation function	ReLU

each step has consistent processing layers. This approach allows the LSTM model to determine the appropriate context for each output step, with dense layers processing each time step independently yet using shared weights. The repeat vector layer adapts the encoder's 2D output to the 3D input format required by the decoder by duplicating the 2D output across multiple time steps.

This process ensures compatibility between the encoder and decoder, enabling effective sequential decoding. This architecture enables the LSTM to effectively capture long-term dependencies in the data sequence while maintaining efficient and consistent feature interpretation. Finally, the output of the LSTM sequence layer is passed through a fully connected layer, where a SoftMax activation function is applied for multi-label classification. The architectural layers of the proposed hybrid deep learning algorithm are illustrated in Figure 5.

4 System Implementation and Simulation Result

In this study, a comprehensive set of fault simulations was conducted, incorporating both symmetrical and unsymmetrical fault types. These faults were applied at various locations along the transmission line, with fault locations ranging from 10% to 90% of the line's total length. Additionally, varying fault resistances were simulated to test the robustness of the proposed models under different operating conditions. The collected data from these simulations were then processed using three distinct classifiers: fault classification type (FCT), line faulty detection (LFD), and fault location estimation (FLE), simultaneously. Furthermore, to ensure effective model training, a custom data split was applied to the dataset for each fault scenario, allocating 70% for training, 10% for validation, and 20% for testing. The training process was carried out over 100 epochs with an adaptive learning rate that decayed every 20 epochs, utilizing the Adam optimizer for efficient weight updates. The models were evaluated based on their ability to classify fault types, detect fault locations,

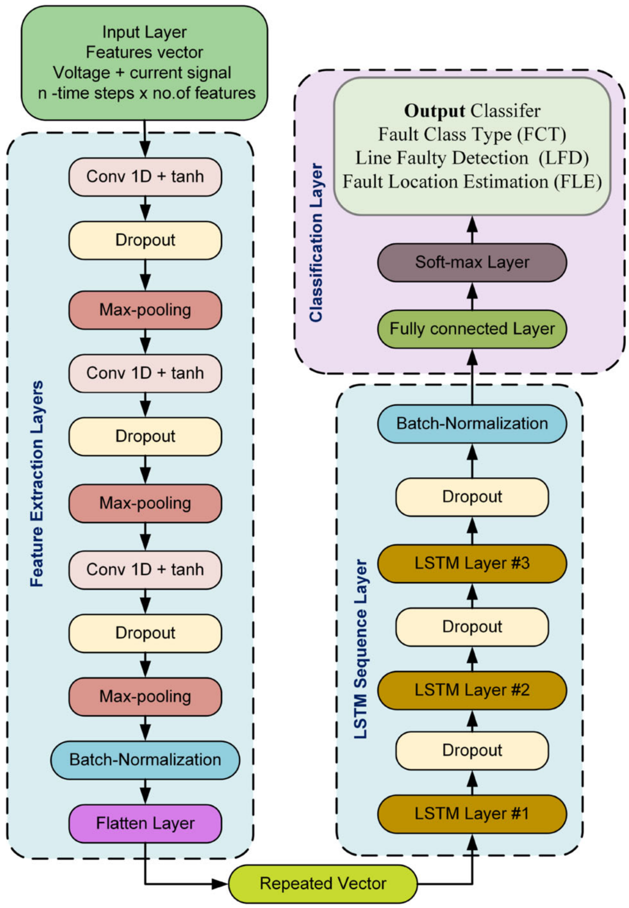

Fig. 5 Detailed layer architecture of the proposed hybrid CNN-LSTM model

and estimate fault locations accurately across a range of simulated scenarios [46–48].

The hyperparameter configurations used during training, including batch size, learning rate, and dropout rates, are summarized in Table 3. The results obtained from these simulations demonstrate the effectiveness of the proposed hybrid CNN-LSTM model in dealing with various fault types and scenarios, showing promising improvements in classification and fault location estimation accuracy compared to existing methods.

In addition, the performance of the proposed model was assessed using multiple evaluation metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's classification effectiveness and involve its overall accuracy, ability to classify relevant instances, and balance in mitigating both false positives and false negatives. Such measures are particularly advantageous in applications where the consequences of classification errors will be changed significantly.

The recall is known as true positive rate (TPR), which measures the true positives (TP) versus the sum of predicted true positives (TP) and predicted false negatives (FN). Furthermore, precision is called the positive predictive value, quantifying the proportion of true positive predictions out of the total predicted positives, which is calculated as the ratio of true positives (TP) to the sum of true positives (TP) and false positives (FP). This metric is particularly valuable in scenarios where minimizing false positives is crucial. In addition, F1-Score represented by the harmonic mean of precision and recall, provides a more balanced evaluation of model performance than accuracy, particularly in the presence of class imbalances. It is especially useful for achieving a balance between precision and recall, making it a valuable preferred metric in real-world scenarios where actual negatives significantly outnumber actual positives. The computation equation for the above terminology is as shown:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$F1 - \text{score} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{8}$$

where TP (true positives) refers to correctly identified positive instances. TN (True Negatives) represents correctly identified negative instances. FP (False positives) indicates instances incorrectly classified as positive. FN (False negatives) refers to instances incorrectly classified as negative.

Table 4 presents the performance evaluation of the fault type classifier (FCT), which includes 11 classes: AG, BG, CG, AB, AC, BC, ABG, ACG, BCG, and NO-FAULT. The model achieved high classification accuracy across all classes, with a perfect score of 100% for most classes. However, the AB class achieved 99% accuracy due to a single instance of misclassification, where n-actual represents the actual number of cases belonging to a specific class, and n-classified represents the number of cases classified as belonging to that class.

Figure 6 presents training and validation accuracy (left) and loss (right) for fault classification over 100 epochs. The model rapidly achieves over 98% accuracy, stabilizing near 100%, with minor fluctuations in validation accuracy suggesting slight overfitting. Both training and validation loss decrease sharply, closely aligning throughout training, indicating effective learning and good generalization. Small spikes in validation loss may result from occasional misclassifications or noise. Overall, the model demonstrates strong performance, with minor fluctuations that could be mitigated through regularization techniques.

Figure 7 presents the confusion matrix. The diagonal represents the matching between the actual and the predicted classes which are the correct prediction, while the rest rep-

Table 4 Performance evaluation of proposed hybrid CNN-LSTM for FCT classifier

Class Name	Precision	Recall	F1-Score	Accuracy %	n -actual (Support)	n-classified
AG	1.00	1.00	1.00	100	115	115
BG	1.00	1.00	1.00	100	108	108
CG	1.00	1.00	1.00	100	93	93
AB	1.00	0.99	1.00	99	103	102
AC	0.99	1.00	1.00	100	105	105
BC	1.00	1.00	1.00	100	96	96
ABG	1.00	1.00	1.00	100	105	105
ACG	1.00	1.00	1.00	100	101	101
BCG	1.00	1.00	1.00	100	97	97
ABC	1.00	1.00	1.00	100	104	104
NO-FAULT	1.00	1.00	1.00	100	248	248

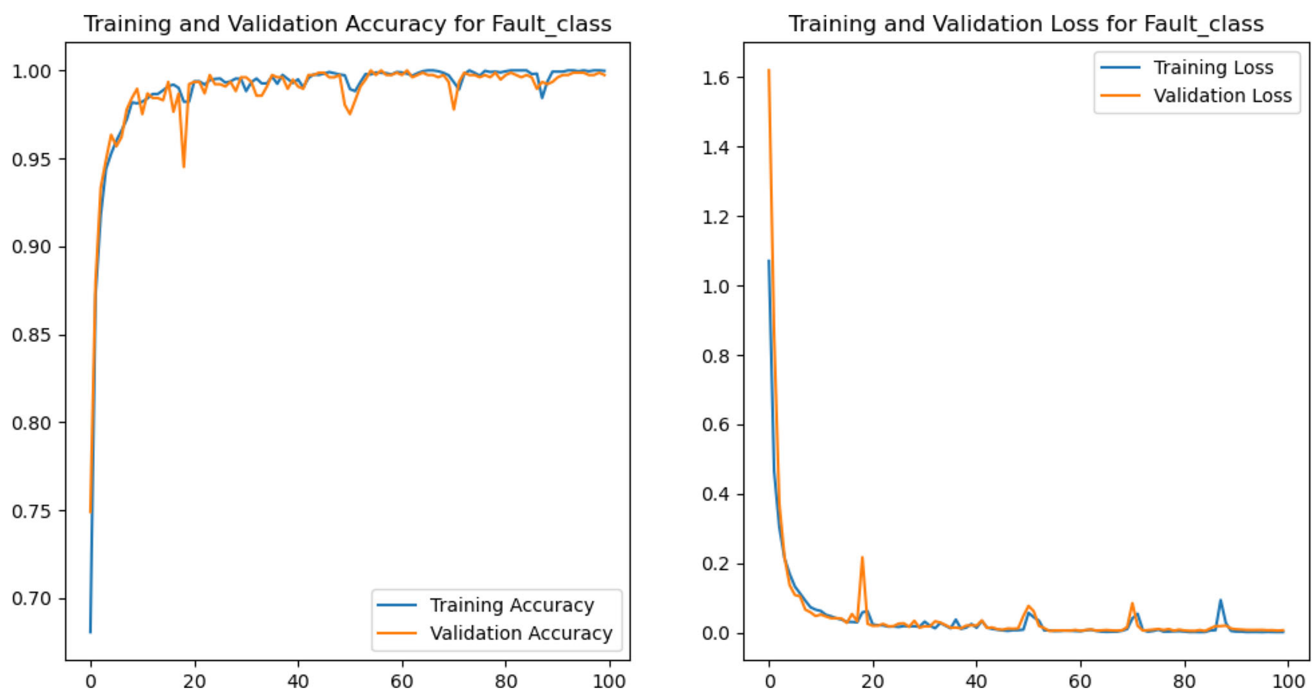

Fig. 6 The loss and accuracy of the training and validation for FCT

resent the misclassified classes. The confusion matrix shows that the model achieves high classification accuracy, with nearly all fault types correctly identified. Predictions are well aligned with true labels, with only one misclassification observed, indicating a minimal error rate. The NO_FAULT class is perfectly classified (248 instances), demonstrating the model's strong ability to distinguish fault and no-fault conditions. Overall, the results suggest a well-generalized model with excellent performance in fault diagnosis.

The cross-validation method with fivefold cross-validation method is employed instead of a custom train/test split, ensuring a more robust evaluation. The dataset is divided into five equal subsets, where four subsets (80%) are used for training, and the remaining one subset (20%) is reserved for testing in each iteration. At the start, Xavier initialization is applied to set the model weights. The training process utilizes the Adam optimizer with an initial learning rate of 0.001, which is adaptively reduced by a factor of 0.1 when learning stagnates. Categorical cross-entropy is chosen as the loss function to optimize classification performance. The model is trained for 50 epochs with a batch size of 16 to ensure effective learning and convergence between training and validation curves. The fivefold cross-validation results of the proposed model are summarized in Table 5. The table shows the fivefold cross-validation results, demonstrating the model's high reliability and consistency in fault classification. With an average accuracy, precision, recall, and F1-score of 0.99596, the model exhibits excellent performance. Minimal variation

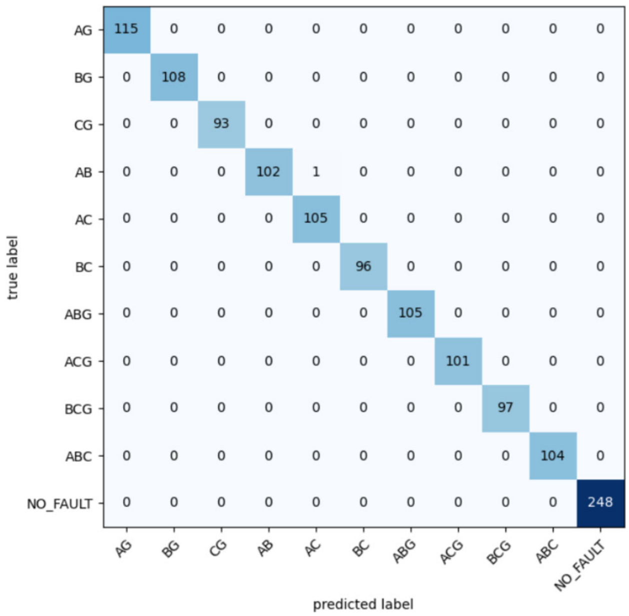

Fig. 7 Confusion matrix for fault class type (FCT)

Table 5 The fivefold cross-validation results of the proposed model for fault classification

Fold No.	Accuracy	Precision	Recall	F1-score
Fold 1	0.9976	0.9976	0.9976	0.9976
Fold 2	0.9952	0.9952	0.9952	0.9952
Fold 3	0.9988	0.9988	0.9988	0.9988
Fold 4	0.9945	0.9945	0.9945	0.9945
Fold 5	0.9937	0.9937	0.9937	0.9937
Average	0.99596	0.99596	0.99596	0.99596

Table 6 Class-wise result of the proposed model for fold no. 5

Class name	Accuracy	Precision	recall	F1-score
AG	0.99	0.98	1.00	0.99
BG	0.99	0.99	1.00	1.00
CG	0.98	1.00	0.98	0.98
AB	1.00	0.99	1.00	1.00
BC	1.00	1.00	0.99	1.00
AC	1.00	1.00	1.00	1.00
ABG	1.00	1.00	1.00	1.00
BCG	1.00	1.00	1.00	1.00
ACG	1.00	1.00	1.00	1.00
ABC	1.00	1.00	1.00	1.00
No-Fault	1.00	1.00	1.00	1.00
Average	0.9964	0.9964	0.9972	0.9973

across folds (accuracy ranging from 0.9937 to 0.9988) highlights its strong generalization ability. The high precision and recall indicate effective fault detection with balanced false positives and false negatives. While Fold 5 shows a slight performance dip, the overall results confirm the model's robustness and effectiveness.

Additionally, Table 6 presents the fivefold cross-validation results, highlighting the model's exceptional performance in fault classification. The consistently high accuracy, precision, recall, and F1-score averaging 0.99596 demonstrate the model's strong reliability and generalization across different data splits. The accuracy ranges from 0.9937 to 0.9988, showing minimal variance, which suggests robustness against dataset variations. The near-perfect precision and recall indicate that the model effectively minimizes false positives and false negatives, ensuring highly reliable fault detection. The slight performance drop in Fold 5 is negligible and does not significantly impact overall effectiveness. These results validate the model's suitability for real-world fault classification applications. Overall, the fivefold cross-validation results reinforce the reliability and effectiveness of the proposed hybrid CNN-LSTM model for fault classification in smart grids.

Table 7 provides an in-depth analysis of the performance metrics for the hybrid CNN-LSTM model when used for fault location estimation (FLE). In this study, the fault location was varied incrementally from 10% to 90% of the line length, with steps of 10%, ensuring that the model's robustness and applicability are demonstrated across different line lengths. This range of fault locations simulates real-world scenarios and ensures the model's generalizability, regardless of the specific characteristics of the power grid.

The fault location estimator (FLE) classifier exhibited strong performance across these various fault locations, achieving consistently high precision, recall, F1-score, and accuracy across all tested line length percentages. The model was able to accurately estimate the fault location in a wide range of scenarios, which underscores the effectiveness of the proposed hybrid CNN-LSTM architecture for this task.

Despite the overall strong performance, slight variations were observed in the results across the different fault location categories. These variations are likely due to the complexities introduced by the diverse fault locations and line lengths, which inherently affect the model's ability to accurately pinpoint the fault location. Specifically, a marginally higher occurrence of false positives (FP) was recorded, which can be attributed to the method of fault location estimation based on percentages of the total line length. This introduces a trade-off between precision and recall, where the model errs on the side of predicting faults that may not exist in order to maximize recall. However, these slight trade-offs did not significantly impact the overall classification performance, with the classifier maintaining accuracy within acceptable bounds.

Table 7 Performance evaluation of proposed hybrid CNN-LSTM for FLE classifier

Class Name	Precision	Recall	F1-Score	Accuracy	n-actual (Support)	n-classified
10% Line Length	1.00	0.97	0.98	98.95	152	147
20% Line Length	1.00	0.95	0.97	98.49	142	135
30% Line Length	0.96	0.95	0.96	98.89	135	128
40% Line Length	0.98	0.93	0.96	99.08	134	125
50% Line Length	0.83	1.00	0.91	99.08	129	129
60% Line Length	0.93	0.98	0.95	98.95	128	125
70% Line Length	0.95	0.92	0.93	99.08	137	126
80% Line Length	0.94	0.94	0.94	98.03	145	137
90 % Line Length	0.99	0.94	0.96	97.18	155	146
No-Fault	1.00	1.00	1.00	100	270	270

The confusion matrix of the FLE classifier, as shown in Fig. 8, reveals additional insights into the model's performance. Notably, the 50% fault location class exhibited a higher number of false negatives (FN) compared to other classes, which is indicative of challenges in accurately estimating fault locations near the end of the line. On the other hand, the no-fault class achieved 100% accuracy, highlighting the model's ability to correctly classify instances where no-fault was present.

These results demonstrate that the hybrid CNN-LSTM model is well suited for fault location estimation, with robust performance across varying fault locations and line lengths. The slight trade-offs between precision and recall are a natural consequence of the model's approach and can be mitigated with further fine-tuning of the classification thresholds. Overall, the model provides a reliable and efficient solution for fault location estimation in power systems Fig. 9.

For the line faulty detection task, the IEEE 14-bus system is composed of 17 transmission lines, each of which is associated with a distinct class in the proposed line faulty detection (LFD) model. This design ensures that each transmission line is treated as a separate entity, which facilitates precise fault identification and enables the classifier to distinguish between faults occurring on different lines. The system's design provides an efficient way to detect faults by assigning each transmission line its own class label, allowing for detailed fault detection and localization.

The performance of the LFD classifier is evaluated across various metrics, which are summarized in Table 8. The table presents the performance metrics of a classification model, showing consistently high precision, recall, and F1-scores (mostly 1.00), indicating strong model performance across various class pairs. Accuracy is also high, with values of 99.59% to 100%, suggesting minimal misclassifications. The "n-classified" values align, indicating that all instances were correctly classified. Overall, the model performs excellently, with balanced classification and reliable results. However, it would be useful to confirm if this high performance extends

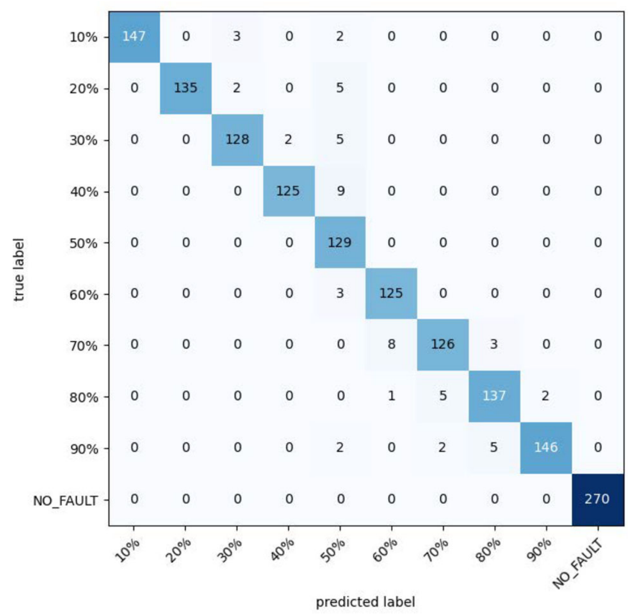

Fig. 8 Confusion matrix for fault location estimator (FLE)

to real-world data or cases with potential class imbalance and noise.

Furthermore, the model exhibits a balanced classification across all 17 classes, which ensures that no single line is unfairly prioritized or overlooked. This comprehensive performance ensures the proposed LFD classifier is well suited for practical deployment in power systems, contributing to efficient fault detection and improved system reliability.

The training and loss curves of the proposed model are illustrated in Fig. 10. Upon analysis, it is observed that the training accuracy begins to converge with the validation accuracy at approximately epoch 50, indicating that the model starts to generalize effectively to unseen data. Furthermore, the loss and validation loss curves demonstrate a consistent decline, reaching their minimum values around epoch 80,

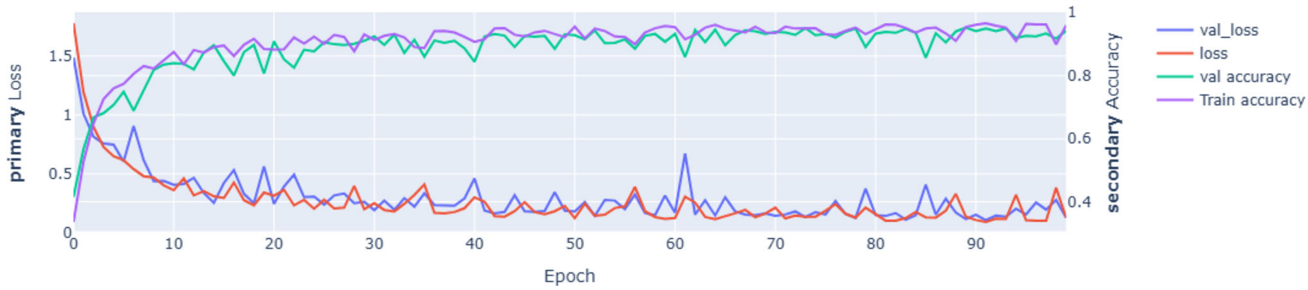

Fig. 9 Training and validation curve of hybrid model for FLE classifier

Table 8 Performance evaluation of proposed hybrid CNN-LSTM for LFD classifier

Class name	Precision	Recall	F1-Score	Accuracy %	n-actual (support)	n-classified
Line1-2	0.93	1.00	0.97	99.59	76	71
Line1-5	1.00	1.00	1.00	100	62	62
Line2-3	1.00	1.00	1.00	100	69	69
Line2-4	1.00	1.00	1.00	100	66	66
Line2-5	1.00	1.00	1.00	100	75	75
Line3-4	1.00	0.95	0.97	99.59	91	91
Line4-5	1.00	1.00	1.00	100	65	65
Line6-12	1.00	1.00	1.00	100	73	73
Line6-13	1.00	1.00	1.00	100	83	83
Line7-8	1.00	1.00	1.00	100	74	74
Line7-9	1.00	1.00	1.00	100	75	75
Line9-10	1.00	1.00	1.00	100	76	76
Line9-14	1.00	1.00	1.00	100	79	79
Line10-11	1.00	1.00	1.00	100	66	66
Line10-14	1.00	1.00	1.00	100	59	59
Line12-13	1.00	1.00	1.00	100	72	72
Line13-14	1.00	1.00	1.00	100	63	63

highlighting the stability and efficiency of the training process.

In addition to the convergence analysis, the classification performance of the line faulty detection (LFD) classifier is further evaluated using the confusion matrix, as shown in Fig. 11. The confusion matrix for the line fault detection (LFD) classifier shows strong classification performance, with most predictions aligning perfectly with the true labels. The diagonal dominance, where high values appear along the diagonal, indicates that the classifier correctly identifies faults with minimal misclassification. Only a few off-diagonal values suggest occasional misclassification, such as in the case of Line1-2, which has a small number of instances predicted as another class. This minimal confusion implies the model is highly effective in distinguishing different fault types. Overall, the classifier demonstrates strong accuracy and reliability in fault detection.

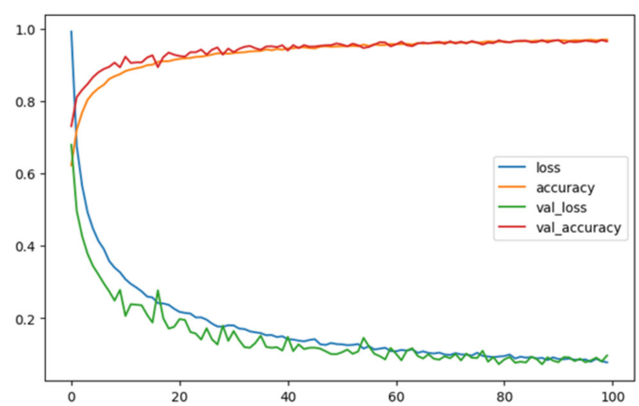

Fig. 10 Training and Validation curve of LFD classifier

5 Discussion

To evaluate the performance of the proposed model in fault classification, faulty section identification, and fault location detection, various configurations were explored by altering

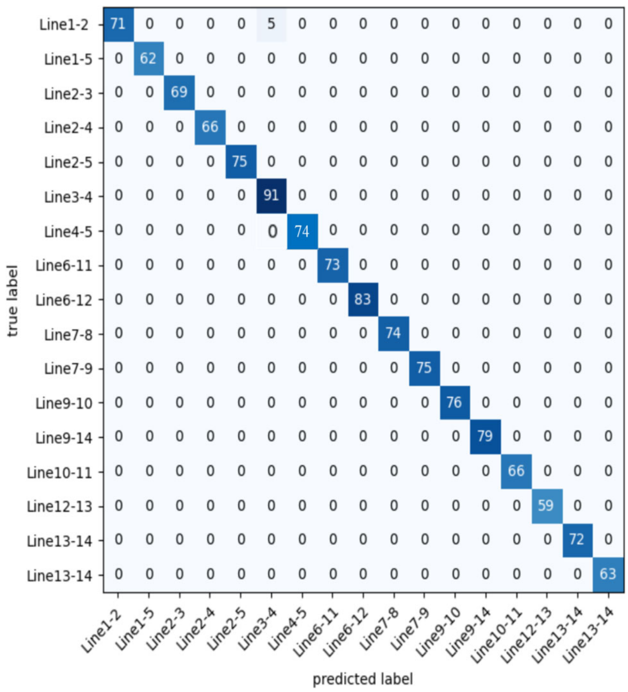

Fig. 11 Confusion matrix for of LFD classifier

the number of convolutional layers. A comparative analysis of the proposed hybrid deep learning models against state-of-the-art approaches is presented in Table 9. All experiments were conducted using the same train/test split and within a consistent computing environment. The results demonstrate that the use of stacked convolutional layers in the proposed model yields superior performance while maintaining a lower number of trainable parameters. Consequently, the proposed model effectively balances performance and computational complexity in terms of trainable parameters.

To further demonstrate the superiority and advancements of the proposed deep learning model, a comparison was conducted against existing fault detection and classification methods. Table 9 presents a comparative analysis of the proposed CNN-LSTM model with various existing fault classification methods used in power systems. The comparison highlights the different grid types, test networks, input data types, classifier types, and whether distributed generation (DG) was included in the study.

The proposed CNN-LSTM model demonstrates exceptional performance with fault classification (FCT), fault location (FLE), and line fault (LFD) detection accuracies of 99.99%, 99.98%, and 96.33%, respectively, on a 14-bus network equipped with DG. These results surpass the most existing methods, as shown in the table. While methods, such as LSTM [15] and CNN [33], achieve high accuracy for FCT and FLE, their LFD detection performance is either absent or significantly lower compared to the proposed model.

Table 9 Comparative analysis with existing research

References	Methods	Data	Grid type	Test network	DG-Equipped	FCT	LFD	FLE
[15]	LSTM	Voltage and Current	Radial	Four-Machine	Yes	97.73	87.5	99.36
[33]	CNN	Voltage and Current	Radial	Two-Machine	Yes	99.92	Nil	99.00
[39]	MCCNN-LSTM	Benchmark Dataset	–	–	No	92.8	Nil	Nil
[49]	SVM	Voltage and Current	Ring	14-bus	No	99.40	Nil	Nil
[50]	SVM/KNN	Voltage	Radial	11-bus	Yes	97.6	Nil	Nil
[51]	ANN	Voltage and Current	Ring/Radial	9-bus	Yes	100.0	Nil	Nil
[52]	ANN	Voltage and Current	Ring	9-bus	Yes	99.85	Nil	Nil
[53]	SAE-DNN	Voltage and Current	Radial	6-bus	Yes	99.52	99.56	Nil
[54]	FCLSTM	Voltage	Radial	Two-Machine	No	97.47	Nil	Nil
[55]	PNN	Current	Radial	Single Machine	Yes	99.30	Nil	Nil
[56]	LSTM RNN	Voltage	Ring Radial	14-bus 39-bus	Yes	98.4 98.0	Nil Nil	90.52 92.24
Proposed CNN-LSTM		Voltage and Current	Ring/Radial	14-bus	Yes	**99.99**	**96.00**	**99.98**

Other approaches, such as MCCNN-LSTM [39], SVM [49], and ANN [51], either lack comprehensive metrics or are limited to specific grid configurations or datasets. Additionally, while hybrid methods, like SAE-DNN [53] and PNN [55], show competitive accuracy for FC or LF, they are often constrained to smaller test networks. The proposed model's ability to handle voltage and current data, combined with its application on a DG-equipped 14-bus system, establishes its superiority in fault classification and location tasks, showcasing its adaptability and robustness across various fault scenarios.

6 Conclusion

Fault occurrence is an inevitable challenge in electrical power networks. Detecting, classifying, and locating faults in smart distribution systems with the presence of distributed generation (DG) resources in real-time is crucial for minimizing downtime, reducing financial losses, and ensuring uninterrupted customer load demand.

This study introduced a novel hybrid deep learning-based intelligent scheme for simultaneous fault type classification (FCT), line faulty detection (FLD), and fault location estimation (FLE). Leveraging recorded voltage and current signals from all buses, the proposed approach utilized a hybrid CNN-LSTM model to overcome the limitations of individual algorithms. While CNN excels in extracting meaningful features from time-series data, its deficiency in capturing long-term dependencies was addressed by integrating LSTM, which specializes in classifying such dependencies.

The data collected from the IEEE 14-bus system was used to train the hybrid model. CNN was employed for feature extraction, and the extracted features were subsequently fed into the LSTM classifier to capture long-term dependencies effectively. Fault simulations were The results demonstrated the superiority of the proposed algorithm, achieving fault type classification accuracy of 99.99%, faulty section identification accuracy of 99.98%, and fault location estimation accuracy of 96.33%.

This innovative method enhances network reliability by enabling rapid and precise fault detection and localization. It is also cost-effective, as it eliminates reliance on communication utilities and wireless sensors, which are prone to failure in traditional networks. By avoiding prolonged fault location and repair times caused by equipment failures, the proposed scheme significantly reduces financial losses and enhances customer satisfaction.

Future Work

The scalability of the proposed CNN-LSTM model is crucial for its applicability in large-scale power networks. While the current study validates its performance on the IEEE 14-bus system, real-world power grids demand greater computational efficiency and adaptability. Future research will focus on extending the model to larger and more complex networks to assess its robustness in diverse operational conditions.

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Author Contributions Ahmed Alhanaf was responsible for conceptualization, data curation, methodology, software, validation, visualization, original draft writing, and review and editing. Murtaza Farsadi and Hasan Huseyin Balik contributed to conceptualization, formal analysis, funding acquisition, investigation, project administration, resource management, supervision, and review and editing.

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Declarations

Conflict of interest No potential Conflict of interest was reported by the author(s) for this research work.

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